

Theta correspondence and Siegel-Weil formula

We mainly follow [Gan22], [Li23] and [Li24].

1 Preliminaries

1.1 Automorphic forms

Let

- k be a number field and $\mathbb{A}_k = \prod'_v k_v$ its ring of adeles which is a restricted product, i.e. consisting of tuples (a_v) where $a_v \in \mathcal{O}_v$ for all but finitely many places
- G be a connected reductive linear algebraic group over k (for simplicity we can assume semi-simple)
- $G(\mathbb{A}_k)$ the adelic points of G . Given any faithful representation $\rho : G \hookrightarrow \mathrm{GL}_n$ we obtain a system of open compact subgroups $K_v = \rho^{-1}(\mathrm{GL}(\mathcal{O}_v)) \subset G(k_v)$ that are hyperspecial for almost all v . We can then have that $G(\mathbb{A}_k) \cong \prod'_v G(k_v)$ where the restricted product is taken with regard to $\{K_v\}$.
- $[G] := G(k) \backslash G(\mathbb{A}_k)$ the automorphic quotient

An *automorphic form* is a function

$$f : [G] \longrightarrow \mathbb{C}$$

satisfying

- (i) f is smooth
- (ii) f is right K_f finite, where $K_f = \prod_{v < \infty} K_v$
- (iii) f is of uniform moderate growth
- (iv) f is $Z(\mathfrak{g})$ -finite, where $\mathfrak{g}$ is the Lie algebra of G .

We denote the space of automorphic forms by $\mathcal{A}(G)$. We have a $G(\mathbb{A}_k)$ -action on $\mathcal{A}(G)$ by right translation.

An automorphic representation π is a representation of $G(\mathbb{A}_k)$ isomorphic to some irreducible subquotient of the $\mathcal{A}(G)$ -module $\mathcal{A}(G)$. As an irreducible abstract representation we have that π decomposes as

$$\pi = \bigotimes_v^I \pi_v$$

where π_v is an irreducible smooth representation. The restricted tensor product guarantees that $\pi_v^{K_v} \neq 0$ for almost all v . In that case we say that π_v is K_v -unramified.

1.2 Reductive dual pairs

Let F be a field of characteristic zero and E/F a quadratic extension. Let W be a symplectic vector space over F . A *reductive dual pair* is a pair of subgroups (G, H) of $\mathrm{Sp}(W)$ such that G is the centralizer of H and H is the centralizer of G in $\mathrm{Sp}(W)$, and both groups act reductively on W . Examples for $F = \mathbb{R}$ include

$$\begin{array}{lll} G = \mathrm{Sp}(2n, \mathbb{R}) & H = O(m, \mathbb{R}) & W = \mathrm{Sp}(2nm, \mathbb{R}) \\ G = U(p, q) & H = U(r, s) & W = \mathrm{Sp}(4(p+q)(r+s), \mathbb{R}). \end{array}$$

We are interested in the following case. Let V be a finite-dimensional Hermitian space over E and W a finite-dimensional skew-Hermitian vector space over E . Then $V \otimes_E W$ is a skew-Hermitian space over E and after restriction of scalars, we can consider it as a symplectic vector space over F with symplectic form $\mathrm{Tr}(\langle -, - \rangle_V \otimes \langle -, - \rangle_W)$. We then have a natural map

$$\iota : U(V) \times U(W) \longrightarrow \mathrm{Sp}(V \otimes_E W)$$

which is injective on each factor. The pair $(U(V), U(W))$ then forms a reductive dual pair.

2 Local theta correspondence

This part mainly follows [Gan22].

We first take a look at the local construction. Let F be a local field, E/F a quadratic extension of local fields. Given a symplectic vector space W and a reductive dual pair (G, H) we are interested in lifting irreducible representations of G to representations of H . Since we have

$$G \times H \longrightarrow \mathrm{Sp}(W)$$

the idea is to take a representation of $\mathrm{Sp}(W)$ and pull it back to $G \times H$. The representation we choose will be the Weil representation which is a representation of a cover of $\mathrm{Sp}(W)$, which we will construct now.

2.1 The Weil representation

Let W be a symplectic vector space over F . Let $H(W)$ denote the *Heisenberg group*, given by

$$H(W) = W \oplus F$$

equipped with a group law

$$(w_1, t_1) \cdot (w_2, t_2) = (w_1 + w_2, t_1 + t_2 + \frac{1}{2}\langle w_1, w_2 \rangle).$$

Note that the center of this group is given by $Z(H(W)) = F$.

The *Stone-von-Neumann theorem* tells us that for a given additive character $\psi : F \rightarrow \mathbb{C}^*$ there is a up to isomorphism unique irreducible representation of $H(W)$ with central character ψ . We denote this representation by ω_ψ . The Stone-von-Neumann theorem also tells us ω_ψ is unitary.

We can explicitly construct the representation. Let

$$W = X \oplus Y$$

with X, Y maximal isotropic subspaces. We extend ψ to $H(X) = X + F$ by setting it zero on X . We then obtain an induced representation (compact induction)

$$\omega_\psi = \text{Ind}_{H(X)}^{H(W)} \psi$$

Let $\mathcal{S} := \mathcal{S}(Y)$ the Schwartz functions on Y , then we can realize the representation on $\mathcal{S}$. We also can define a group action of $\text{Sp}(W)$ on $H(W)$ by setting

$$g \cdot (w, t) = (gw, t)$$

then $\text{Sp}(W)$ acts trivially on F and therefore for any $g \in \text{Sp}(W)$ the representation ${}^g\omega_\psi = \omega_\psi \circ g^{-1}$ also has central character ψ . By Stone-von-Neumann there exists an invertible operator $A_\psi(g)$ on $\mathcal{S}$ such that

$$A_\psi(g) \circ {}^g\omega_\psi(h) = \omega_\psi(h) \circ A_\psi(g)$$

for all $h \in H(W)$. By Schur's lemma, we have that $A_\psi(g)$ is defined up to multiplication by $\mathbb{C}^*$. Since ω_ψ is unitary, we require $A_\psi(g)$ to be unitary, then $A_\psi(g)$ is defined up to $S^1 \subset \mathbb{C}^*$. Therefore we get

$$A_\psi : \text{Sp}(W) \rightarrow \text{GL}(\mathcal{S})/S^1$$

We define the *metaplectic group* as

$$\text{Mp}(W) := \{(g, T) \in \text{Sp}(W) \times \text{GL}_n(\mathcal{S}) \mid T(g) \circ {}^g\omega_\psi(h) = \omega_\psi(h) \circ T(g) \text{ for all } h \in H(W)\}$$

then it fits in the following diagram

$$\begin{array}{ccccccc}
 1 & \longrightarrow & S^1 & \longrightarrow & \mathrm{Mp}(W) & \xrightarrow{\mathrm{pr}_{\mathrm{Sp}(W)}} & \mathrm{Sp}(W) \longrightarrow 1 \\
 & & \parallel & & \downarrow \mathrm{pr}_{\mathrm{GL}(\mathcal{S})} & & \downarrow A_\psi \\
 1 & \longrightarrow & S^1 & \longrightarrow & \mathrm{GL}(\mathcal{S}) & \longrightarrow & \mathrm{GL}(\mathcal{S})/S^1 \longrightarrow 1
 \end{array}$$

which gives us a representation $\mathrm{Mp}(W) \longrightarrow \mathrm{GL}(\mathcal{S})$, which we will call the *Weil representation*, also denoted by ω_ψ .

Note that although the construction depends on the choice of ψ , the class of groups extensions of $\mathrm{Mp}(W)$ does not depend on the choice of ψ , which is why we will just speak about the metaplectic group.

2.2 Local theta lifts

We originally planned to pull back a representation of $\mathrm{Sp}(W \otimes_E V)$ along

$$\iota : U(V) \times U(W) \longrightarrow \mathrm{Sp}(V \otimes_E W).$$

Given that the Weil representation is a representation of $\mathrm{Mp}(V \otimes_E W)$, we need to lift ι to a map

$$\tilde{\iota} : U(V) \times U(W) \longrightarrow \mathrm{Mp}(V \otimes_E W).$$

such a lifting can be specified by picking characters χ_V, χ_W of E^* such that

$$\chi_V|_{F^*} = \eta_{E/F}^{\dim V} \quad \chi_W|_{F^*} = \eta_{E/F}^{\dim W}$$

where $\eta_{E/F} : F^* \longrightarrow \{\pm 1\} \subset \mathbb{C}^*$ is given by local class field theory. We don't go further into detail here, but the liftings were constructed by Kudla in [Kud86].

We can now pull back the Weil representation to get

$$\Omega = \tilde{\iota}^*(\omega_\psi).$$

Note that Ω depends on the choice of ψ, χ_V and χ_W . If we write $\mathrm{Irr}(U(V))$ for the set of equivalence classes of irreducible smooth representations on $U(V)$ we a correspondence between $\mathrm{Irr}(U(V))$ and $\mathrm{Irr}(U(W))$ via the set

$$\{(\pi, \sigma) \in \mathrm{Irr}(U(V)) \times \mathrm{Irr}(U(W)) \mid \pi \otimes \sigma \text{ is an irreducible quotient of } \Omega\}.$$

Given $\pi \in \mathrm{Irr}(U(V))$ one may look at the maximal π -isotypic quotient

$$\Omega / \bigcap_{f \in \mathrm{Hom}(\Omega, \pi)} \ker(f)$$

which is a $U(V) \times U(W)$ quotient of the form

$$\pi \otimes \Theta(\pi)$$

for some smooth representation $\Theta(\pi)$ of $U(W)$. Note that we have

$$\Theta(\pi) = (\Omega \otimes \pi^\vee)_{U(V)}.$$

We will call the irreducible quotients of $\Theta(\pi)$ the local theta lifts of π .

We summarize some results about the big theta lift.

Proposition 2.1. For any $\pi \in \text{Irr}(U(V))$, $\Theta(\pi)$ is of finite length. In particular, if $\Theta(\pi)$ is nonzero, it has finitely many irreducible quotients and we denote its maximal semisimple quotient by $\theta(\pi)$.

Theorem 2.2 (Howe's duality theorem, [GT15]). (i) If $\Theta(\pi)$ is nonzero, then it has a unique irreducible quotient. In other words, $\theta(\pi)$ is irreducible or 0.

(ii) If $\theta(\pi) \cong \theta(\pi') \neq 0$ then $\pi \cong \pi'$.

The first question one might ask is when a theta lift of a given irreducible representation is non-zero.

3 Global theta correspondence

We will now turn to the global case, where k is a number field and E/k a quadratic extension. Let V be a finite-dimensional Hermitian vector space over E and W a finite-dimensional skew-Hermitian vector space over E . Just like before, we want to consider a reductive dual pair (G, H) with $G = U(V)$ and $H = U(W)$ with a map

$$\iota : G \times H \longrightarrow \text{Sp}(V \otimes_E W).$$

Using the Weil representation, we now would like to lift automorphic forms of G to H . In order to do that, we will need to look at adelic points of the given groups. However, since the metaplectic group is not an algebraic group, we will first need to construct its adelic points.

3.1 Adelic points of the metaplectic group

Fix an additive character

$$\psi : \mathbb{A}_k \longrightarrow \mathbb{C}^*$$

which the for each place v determines a Weil representation on the metaplectic group

$$1 \longrightarrow S^1 \longrightarrow \mathrm{Mp}((V \otimes_E W)_v) \longrightarrow \mathrm{Sp}((V \otimes_E W)_v) \longrightarrow 1.$$

If we fix a family of hyperspecial maximal compact subgroups K_v we can take

$$\mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k) = \prod_v' \mathrm{Mp}((V \otimes_E W)_v)/Z$$

where Z is the central subgroup $\{(z_v)_v \in \bigoplus_v S^1 \mid \prod_v z_v = 1\}$. We obtain

$$1 \longrightarrow S^1 \longrightarrow \mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k) \longrightarrow \mathrm{Sp}(V \otimes_E W)(\mathbb{A}_k) \longrightarrow 1$$

and one can construct a splitting from the subgroup $\mathrm{Sp}(V \otimes_E W)(k)$ into $\mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k)$, so we will consider it as a subgroup of $\mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k)$. We can now consider the Weil representation

$$\omega_\psi = \bigotimes_v' \omega_{\psi_v}$$

on $\prod_v' \mathrm{Mp}((V \otimes_E W)_v)$ which factors to a representation of $\mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k)$. The Weil representation can be realized on the space $\mathcal{S}(Y(\mathbb{A}_k)) = \bigotimes_v' \mathcal{S}(Y_v)$ where Y_v is chosen as for the local Weil representation.

3.2 Global theta lifts

Let $G = U(V)$ and $H = U(W)$. Just like before we want to consider a lift of

$$\iota : G(\mathbb{A}_k) \times H(\mathbb{A}_k) \longrightarrow \mathrm{Sp}(V \otimes_E W)(\mathbb{A}_k).$$

to a map

$$\tilde{\iota} : G(\mathbb{A}_k) \times H(\mathbb{A}_k) \longrightarrow \mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k).$$

by picking characters χ_V, χ_W of E^* such that

$$\chi_V|_{\mathbb{A}_k^*} = \eta_{E/F}^{\dim V} \quad \chi_W|_{\mathbb{A}_k^*} = \eta_{E/F}^{\dim W}$$

where $\eta_{E/F} : \mathbb{A}_k^* \longrightarrow \{\pm 1\} \subset \mathbb{C}^*$ is given by local class field theory. One can show that this map maps $G(k) \times H(k)$ into $\mathrm{Sp}(V \otimes_E W)(k) \subset \mathrm{Mp}(V \otimes_E W)(\mathbb{A}_k)$. Pulling back the Weil representation via $\tilde{\iota}$ gives us a Weil representation ω on $G(\mathbb{A}_k) \times H(\mathbb{A}_k)$ realized on $\mathcal{S}(Y(\mathbb{A}_k))$.

We can now define the automorphic theta distribution

$$\begin{aligned} \theta : \mathcal{S}(Y(\mathbb{A}_k)) &\longrightarrow \mathcal{A}(G \times H) \\ \theta(\varphi)(g, h) &= \sum_{y \in Y(k)} (\omega(g, h)\varphi)(y) \end{aligned}$$

which is well-defined since one can show that $\theta(\varphi)(-, -)$ as a function on $G(\mathbb{A}_k) \times H(\mathbb{A}_k)$ is invariant under $G(k) \times H(k)$ using Poisson summation. This distribution is also equivariant under right $G(\mathbb{A}_k) \times H(\mathbb{A}_k)$ -action. We will generally write $\theta(g, h, \varphi)$ for $\theta(\varphi)(g, h)$.

Using $\theta(g, h, \varphi)$ as an integral kernel, we can now define the theta lifts of an automorphic form $f \in \mathcal{A}(G)$ given by

$$\theta_\varphi(f)(h) = \langle \theta(-, h, \varphi), f \rangle = \int_{[G]} \theta(g, h, \varphi) \overline{f(g)} dg$$

where dg denotes the Tamagawa measure. If f is cuspidal, the integral is absolutely convergent and $\theta_\varphi(f)$ defines an automorphic form on H .

Finally, for a given cuspidal automorphic representation π of G we can consider the $G(\mathbb{A}_k) \times H(\mathbb{A}_k)$ -equivariant map

$$\begin{aligned} \theta : \mathcal{S}(Y(\mathbb{A}_k)) \otimes \pi^\vee &\longrightarrow \mathcal{A}(H) \\ (\varphi, f^\vee) &\longmapsto \theta_\varphi(f). \end{aligned}$$

and let $\Theta(\pi)$ denote the image of this map.

Finally we want to relate the global and local theta lifts.

Proposition 3.1 ([Gan22]). Suppose that $\Theta(\pi)$ is non-zero and is contained in the space $\mathcal{A}_2(H)$ of square-integrable automorphic forms on H . Then

$$\Theta(\pi) \cong \bigotimes_v \theta(\pi_v).$$

4 The Siegel-Weil formula

This part mainly follows [Li24] and [Li23].

As before, let k be a number field, and let E/k be a quadratic extension. We now want to state a formula that relates the theta lifts to Eisenstein series. For this section we specify $W = E^{2n}$ with skew-Hermitian form given by $\begin{pmatrix} 0 & id \\ -id & 0 \end{pmatrix}$. Let $m = \dim V$. We let $G = U(W)$ and $H = U(V)$. Note that in this case, the Weil representation can be realized on the space $\mathcal{S}(V(\mathbb{A}_k)^n)$. We define the *theta integral* to be

$$I(g, \varphi) = \int_{[H]} \theta(g, h, \varphi) dh.$$

The integral is absolutely convergent for all φ if and only if the pair (V, W) satisfies Weil's convergence conditions: for α the dimension of the maximal isotropic subspace of V

$$\alpha = 0 \text{ or } \alpha > 0 \text{ and } m - \alpha > n$$

in this case $I(-, \theta)$ is an automorphic form on G , the lift of the constant automorphic form on H . We have a $G(\mathbb{A}_k)$ -equivariant distribution

$$\begin{aligned}\mathcal{S}(V(\mathbb{A}_k)^n) &\longrightarrow \mathcal{A}(G) \\ \varphi &\longmapsto I(-, \varphi).\end{aligned}$$

We now define the Siegel Eisenstein series. For $s \in \mathbb{C}$ we have a distribution

$$\begin{aligned}\mathcal{S}(V(\mathbb{A}_k)^n) &\longrightarrow \text{Ind}_{P(\mathbb{A}_k)}^{G(\mathbb{A}_k)}(\chi| \cdot |_F^{s+\frac{n}{2}}) \\ \varphi &\longmapsto \Phi_\varphi(-, s).\end{aligned}$$

where

$$\Phi_\varphi(g, s) := (\omega(g)\varphi)(0) \cdot |\det a(g)|_F^{s-s_0}$$

with $s_0 := \frac{m-n}{2}$ and $a(g) = a$ is such that $g = kan$ from the Iwasawa decomposition. This distribution is $G(\mathbb{A}_k)$ -equivariant if and only if $s = s_0$. We define the *Siegel Eisenstein series* to be

$$E(g, s, \varphi) = \sum_{\gamma \in P(k) \backslash G(k)} \Phi_\varphi(\gamma g, s)$$

which converges for $\text{Re } s > \frac{n}{2}$ and we can show it has meromorphic continuation. If at $s = s_0$ it is holomorphic we have the following distribution

$$\begin{aligned}\mathcal{S}(V(\mathbb{A}_k)^n) &\longrightarrow \mathcal{A}(G) \\ \varphi &\longmapsto E(-, s_0, \varphi).\end{aligned}$$

Theorem 4.1. Assume that the pair (V, W) satisfies Weil's convergence condition. Then $E(g, s, \varphi)$ is holomorphic at s_0 and

$$\kappa \cdot I(g, \varphi) = E(g, s_0, \varphi)$$

where $\kappa = \frac{1}{2}$ if $m > n$ and $\kappa = 1$ otherwise.

5 The Doubling Integral

This part mainly follows [Li24].

We spent a few sentences on the doubling method which will be used to prove the Rallis inner product formula next week. As before, let k be a number field, and let E/k be a quadratic extension. Let W as before and $W^\square = W \oplus (-W)$ of dimension $4n$ over k . Let $G^\square = U(W^\square)$. We obtain a Weil representation $\omega^\square$ of $G^\square(\mathbb{A}_k)$ on

$\mathcal{S}(V(\mathbb{A}_k)^{2n})$. If ω denotes the usual Weil representation of $G(\mathbb{A}_k)$ then we have an isomorphism $G(\mathbb{A}_k) \times G(\mathbb{A}_k)$ representations

$$\delta : \omega \boxtimes (\omega^\vee \otimes \chi) \cong \omega^\square|_{G(\mathbb{A}_k) \times G(\mathbb{A}_k)}.$$

For any $\varphi_1, \varphi_2 \in \mathcal{S}(V(\mathbb{A}_k)^n)$, we have a Siegel-Eisenstein series $E(g, s, \delta(\varphi_1 \otimes \bar{\varphi}_2))$ on $G^\square$. Let π be a cuspidal automorphic representation of G . For any $f_1, f_2 \in \pi$, define the global doubling zeta integral

$$Z(s, f_1, f_2, \varphi_1, \varphi_2) = \int_{[G] \times [G]} \overline{f_1(g_1)} f_2(g_2) \cdot E((g_1, g_2), s, \delta(\varphi_1 \otimes \bar{\varphi}_2)) \chi^{-1}(\det g_2) dg_1 dg_2$$

which converges for $\operatorname{Re}(s) > 0$ and has meromorphic continuation on $\mathbb{C}$. If f_i and φ_i are factorizable we can decompose the doubling integral into local zeta doubling integrals

$$Z(s, f_1, f_2, \varphi_1, \varphi_2) = \prod_v Z_v(s, f_{1,v}, f_{2,v}, \varphi_{1,v}, \varphi_{2,v}).$$

with

$$Z_v(s, f_{1,v}, f_{2,v}, \varphi_{1,v}, \varphi_{2,v}) = \int_{G(k_v)} \overline{\langle g_v f_{1,v}, f_{2,v} \rangle_{\pi_v}} \cdot \Phi_{\varphi_{1,v} \otimes \bar{\varphi}_{2,v}}((g_v, 1), s) dg_v$$

When all the data are normalized unramified at a finite place v with $\langle f_{1,v}, f_{2,v} \rangle = 1$ we have

$$Z_v(s, f_{1,v}, f_{2,v}, \varphi_{1,v}, \varphi_{2,v}) = \frac{L(s + \frac{1}{2}, \pi_v \times \chi)}{b_{2n,v}(s)}$$

where $L(s + \frac{1}{2}, \pi_v \times \chi_v)$ is the doubling L -factor and $b_{k,v}(s)$ is a product of Hecke L -factors.